Name of Course	**Bachelor of Computer Application (BCA)**
Semester	**V**
Name of Subject	**Computer Network**
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Lectures	**50 Lectures**

Unit I

1. Introduction to computer network

1.1 What is Network? , Benefits of Networking

1.2 Wired Transmission media – Magnetic media, Twisted Pair, Coaxial Cable, Fiber Optics

1.3 Wireless Transmission media – The Electromagnetic Spectrum, Radio Transmission, Microwave Transmission, Infrared

1.4 Topologies with advantages & disadvantages:-Bus, Ring, Star, Tree, Mesh

1.5 Types of Networks – LAN, MAN, WAN

1.1 What is Network? , Benefits of Networking:

What is a Network?

A **network** is a group of two or more **interconnected devices or systems** that can share data, resources, and information. These devices communicate using **communication protocols** over wired or wireless mediums. Common examples include **computer networks**, **mobile networks**, and the **Internet**.

Benefits of Networking:

1. Resource Sharing:

Devices like printers, scanners, and files can be shared among multiple users, reducing cost and improving efficiency.

2. Data Sharing and Communication:

Users can easily exchange information through emails, messaging, or file transfers.

3. Centralized Data Management:

Data can be stored, backed up, and managed from a central server, making access and maintenance easier.

4. Improved Security and Access Control:

Networks allow administrators to manage user permissions, apply security policies, and protect sensitive data.

5. Remote Access and Collaboration:

Users can access network resources and collaborate with others from different locations in real-time.

In summary, a network connects devices to enable communication, collaboration, and resource sharing, making it essential in modern computing environments.

1.2 Wired Transmission media – Magnetic media, Twisted Pair, Coaxial Cable, Fiber Optics:

Wired Transmission Media:

Wired transmission media refers to the **physical cables or wires** used to transmit data signals from one device to another.

1. Magnetic Media:

It includes **hard disks, floppy disks, and magnetic tapes**.
Data is stored magnetically and transferred by physically moving the media between devices.
Mostly used for **offline data transfer and backup**.

2. Twisted Pair Cable:

Made of **two insulated copper wires twisted together**.
Commonly used in **telephone lines and LANs (e.g., Ethernet cables)**.
Types: **Unshielded (UTP)** and **Shielded (STP)**.
Low cost and easy to install but has limited bandwidth and more signal interference.

3. Coaxial Cable:

Consists of a **central copper conductor**, **insulating layer**, **metallic shield**, and **outer cover**.
Provides **better shielding** than twisted pair.
Used in **cable TV networks and broadband internet**.

4. Fiber Optic Cable:

Transmits data as **light signals** through **glass or plastic fibers**.
Offers **high speed, high bandwidth, and long-distance transmission**.
Immune to electromagnetic interference.
Used in **high-speed internet, telecommunication, and backbone networks**.

In summary, each wired media type has different applications based on speed, distance, and cost. Fiber optics offer the fastest and most reliable transmission, while twisted pair is cost-effective for short distances.

1.3 Wireless Transmission media – The Electromagnetic Spectrum, Radio Transmission, Microwave Transmission, Infrared

Wireless Transmission Media:

Wireless transmission media use **electromagnetic waves** to transmit data **without physical wires**. It enables **remote communication** between devices over short or long distances.

1. The Electromagnetic Spectrum:

It refers to the **range of all types of electromagnetic radiation**, including **radio waves, microwaves, infrared, visible light, etc.**

Different parts of the spectrum are used for **different types of wireless communication** depending on frequency and wavelength.

2. Radio Transmission:

Uses **low-frequency radio waves** to transmit signals.
Can **penetrate walls** and travel long distances.
Used in **AM/FM radios, mobile phones, and wireless networking (Wi-Fi)**.
Offers **omnidirectional** communication.

3. Microwave Transmission:

Uses **high-frequency radio waves (1–100 GHz)**.
Requires **line-of-sight** between transmitter and receiver.
Used for **television signals, satellite communication, and long-distance telephone calls**.
Has higher bandwidth than radio waves.

4. Infrared (IR):

Uses **infrared light waves** for very short-range communication.
Cannot pass through walls.
Commonly used in **TV remote controls, short-range data transfer (like file sharing between phones)**.

In summary, wireless media allow flexible and mobile communication using different parts of the electromagnetic spectrum, each suited for specific distances, data rates, and applications.

1.4 Topologies with advantages & disadvantages:-Bus, Ring, Star, Tree, Mesh:

1. Bus Topology:

Structure:

All devices are connected to a single central cable (backbone).

Advantages:

Easy to implement and cost-effective.
Requires less cable than other topologies.

Disadvantages:

Difficult to troubleshoot.
A break in the backbone affects the entire network.
Limited cable length and number of devices.

2. Ring Topology:

Structure:

Devices are connected in a circular fashion, where each device is connected to two others.

Advantages:

Data flows in one direction, reducing data collisions.
Good performance under light loads.

Disadvantages:

Failure of one node can affect the entire network.
Difficult to reconfigure or add new devices.

3. Star Topology:

Structure:

All devices are connected to a central hub or switch.

Advantages:

Easy to install and manage.
Failure in one device doesn't affect others.

Disadvantages:

If the central hub fails, the entire network goes down.
Requires more cable than bus or ring.

4. Tree Topology:

Structure:

A hybrid of star and bus; groups of star-configured networks are connected to a main backbone.

Advantages:

Scalable and easy to manage.
Fault isolation is easier.

Disadvantages:

Backbone failure can bring down the whole system.
Requires more cable and configuration.

5. Mesh Topology:

Structure:

Every device is connected to every other device.

Advantages:

Provides high redundancy and reliability.
Failure of one link does not affect communication.

Disadvantages:

Very costly due to a large number of cables and ports.
Complex to install and maintain.

In summary, each topology suits different needs based on **cost, scalability, fault tolerance**, and **ease of maintenance**.

1.5 Types of Networks – LAN, MAN, WAN:

1. LAN (Local Area Network):

A network that connects **computers within a small geographical area** such as a home, school, or office.
High data transfer speed and **low cost**.
Example: Connecting PCs and printers in a computer lab.

Advantages:

Fast and secure communication.
Easy to share resources like files and printers.

Disadvantages:

Limited to a small area.
Maintenance may be needed if the number of users grows.

2. MAN (Metropolitan Area Network):

Covers a **larger area than LAN**, like a **city or large campus**.
Connects multiple LANs using routers or fiber optic cables.
Example: A network connecting branches of a bank across a city.

Advantages:

Provides high-speed communication across a city.
Useful for organizations with multiple buildings or offices.

Disadvantages:

More expensive than LAN.
Needs skilled management and maintenance.

3. WAN (Wide Area Network):

Covers a **very large area**, such as **countries or continents**.
The **Internet** is the best example of a WAN.
Uses public or leased communication lines like satellites or undersea cables.

Advantages:

Enables long-distance communication and global connectivity.
Supports large-scale business operations.

Disadvantages:

Expensive to set up and maintain.
Lower speed and more security risks compared to LAN/MAN.

In summary, the type of network—**LAN, MAN, or WAN**—depends on the geographical area, number of devices, and communication needs.

Unit II

2. Reference Model and Ethernet

2.1 The need for layered architecture (protocol hierarchies)
2.2 OSI reference Model
2.3 TCP/IP reference Model
2.4 Ethernet Technology - Types of Ethernet, properties of Ethernet, Collision detection and Recovery, Ethernet hardware address, Ethernet Frame Format
2.5 Wireless LAN
2.6 Bluetooth

2.1 The need for layered architecture (protocol hierarchies):

What is Layered Architecture?

Layered architecture, also known as **protocol hierarchy**, is a design approach where the communication process is divided into **separate layers**, each with a specific function. A well-known example is the **OSI (Open Systems Interconnection) model**.

Need for Layered Architecture:

1. Simplifies Network Design:

Dividing the network into layers makes it easier to design, develop, and understand. Each layer performs a specific task.

2. Modularity:

Each layer is independent, so changes in one layer do not affect others. This makes **maintenance and upgrades** easier.

3. Interoperability:

Standard layers allow **different hardware and software** from various vendors to communicate effectively.

4. Troubleshooting and Debugging:

Problems can be **isolated layer by layer**, which simplifies error detection and resolution.

5. Flexibility and Scalability:

New protocols or technologies can be added in specific layers without redesigning the whole system.

In summary, layered architecture is essential for organizing complex network communication into manageable, independent parts that ensure compatibility, scalability, and ease of maintenance.

2.2 OSI reference Model:

What is the OSI Model?

The **OSI (Open Systems Interconnection)** model is a **theoretical framework** developed by ISO (International Organization for Standardization) that defines **how data is transmitted** over a network in **7 distinct layers**. Each layer performs a specific function and communicates with the layers directly above and below it.

7 Layers of OSI Model:

1. Physical Layer:

Deals with **transmission of raw bits** over physical medium (cables, switches).
Example: Ethernet cables, connectors.

2. Data Link Layer:

Responsible for **node-to-node delivery**, **error detection**, and **MAC addressing**.
Example: Switches, MAC address.

3. Network Layer:

Handles **routing** and **logical addressing (IP)** to deliver data across networks.
Example: Routers, IP address.

4. Transport Layer:

Ensures **reliable data transfer**, **flow control**, and **error recovery**.
Example: TCP, UDP.

5. Session Layer:

Manages **sessions** or **connections** between applications.
Controls dialog management.

6. Presentation Layer:

Handles **data translation**, **encryption/decryption**, and **compression**.
Example: Encoding formats like JPEG, ASCII.

7. Application Layer:

Closest to the user; provides **network services** to applications.
Example: HTTP, FTP, SMTP.

In Summary:

The OSI model helps in **standardizing network communication**, making it **easier to develop interoperable hardware/software**, and to **troubleshoot network issues** layer by layer.

2.3 TCP/IP reference Model:

What is the TCP/IP Model?

The **TCP/IP (Transmission Control Protocol/Internet Protocol)** model is a **conceptual framework** used to describe how data is transmitted over the Internet. It is the foundation of the modern **Internet** and has **4 layers**.

Layers of TCP/IP Model:

1. Application Layer:

Provides **network services directly to users and applications**.
Supports protocols like **HTTP, FTP, SMTP, DNS**.
Equivalent to the top three layers of the OSI model (Application, Presentation, Session).

2. Transport Layer:

Responsible for **end-to-end communication**, **error checking**, and **flow control**.
Main protocols:

TCP (Transmission Control Protocol) – reliable connection.

UDP (User Datagram Protocol)** – faster but less reliable.

3. Internet Layer:

Handles **logical addressing (IP addresses)** and **routing of data packets** across networks.
Key protocols: **IP, ICMP, ARP**.

4. Network Access Layer (Link Layer):

Deals with **physical transmission of data** over a network medium.
Includes hardware addressing (MAC), device drivers, and network interface cards (NICs).
Equivalent to the OSI's **Data Link** and **Physical** layers.

In Summary:

The TCP/IP model is a **simplified and practical model** used in real-world networking. It ensures **efficient communication over the Internet**, and serves as the **standard protocol suite** for all internet-based communication.

2.4 Ethernet Technology - Types of Ethernet, properties of Ethernet, Collision detection and Recovery, Ethernet hardware address, Ethernet Frame Format:

1. Types of Ethernet:

Standard Ethernet (10 Mbps) – Also known as **10Base-T**.
Fast Ethernet (100 Mbps) – Known as **100Base-TX**.
Gigabit Ethernet (1 Gbps) – Known as **1000Base-T**.
10-Gigabit Ethernet (10 Gbps) – Used in high-speed backbone networks.

Each type differs in **speed**, **cabling type**, and **usage area**.

2. Properties of Ethernet:

Uses CSMA/CD for managing access to the network.
Broadcast network – All nodes share the same communication channel.
Star topology with switches is common.
Supports scalable data rates from 10 Mbps to 100 Gbps.
Based on the **IEEE 802.3** standard.

3. Collision Detection and Recovery (CSMA/CD):

Carrier Sense Multiple Access with Collision Detection (CSMA/CD) is used.
Before sending data, a device **listens to the channel**.
If two devices transmit at the same time, a **collision** occurs.
Devices detect the collision and send a **jam signal**.
Then, they wait a **random backoff time** before retransmitting.

> Note: CSMA/CD is mainly used in **older half-duplex Ethernet**. Modern Ethernet with switches uses **full duplex** and avoids collisions.

4. Ethernet Hardware Address (MAC Address):

Each Ethernet device has a unique **48-bit MAC (Media Access Control) address**.
Example: `00:1A:2B:3C:4D:5E`.

Used to identify devices at the **Data Link Layer**.

5. Ethernet Frame Format:

Preamble	Destination MAC	Source MAC	Type/Length	Data	CRC

Preamble (7 bytes): Synchronization bits.
Start Frame Delimiter (1 byte): Marks start of frame.
Destination MAC (6 bytes): Receiver's address.
Source MAC (6 bytes): Sender's address.
Type/Length (2 bytes): Indicates payload type.
Data (46–1500 bytes): Actual data.
CRC (4 bytes): Error checking (Cyclic Redundancy Check).

In Summary:

Ethernet is the most widely used LAN technology. It supports various speeds, uses MAC addresses for device identification, manages collisions using CSMA/CD (in older systems), and transmits data using a standardized frame format.

2.5 Wireless LAN:

What is Wireless LAN?

A **Wireless LAN (WLAN)** is a **local area network** that uses **wireless communication (radio waves)** instead of physical cables to connect devices like laptops, smartphones, and printers within a limited area such as a home, office, or campus.

Key Features of WLAN:

1. Uses IEEE 802.11 Standards:

Commonly known as **Wi-Fi**.
Supports various versions like 802.11a/b/g/n/ac/ax.

2. Wireless Access Point (WAP):

Acts as a hub for connecting wireless devices to a wired network or the Internet.

3. SSID (Service Set Identifier):

Unique name used to identify a WLAN.

4. Security Mechanisms:

Includes **WEP, WPA, WPA2, and WPA3** for encrypting wireless communication.

Advantages of WLAN:

Mobility: Users can move freely within the coverage area.
Ease of Installation: No physical cabling is needed.
Scalability: Devices can be easily added to the network.
Cost-Effective: Saves on wiring and maintenance costs.

Disadvantages of WLAN:

Interference: Susceptible to radio interference and signal loss.
Security Risks: Wireless signals can be intercepted if not encrypted properly.
Limited Range: Coverage depends on the strength of the access point.

In Summary:

A Wireless LAN provides **flexible and mobile network access** using radio waves, making it ideal for modern environments that require easy connectivity and minimal wiring.

2.6 Bluetooth:

What is Bluetooth?

Bluetooth is a **wireless communication technology** used for **short-range data exchange** between electronic devices using **radio waves** in the **2.4 GHz ISM band**. It enables the creation of **personal area networks (PANs)**.

Key Features of Bluetooth:

1. Short Range Communication:

Typical range is **10 meters**, extendable up to **100 meters** with enhanced power.

2. Low Power Consumption:

Designed for devices like smartphones, headsets, keyboards, and smartwatches.

3. Device Pairing:

Devices must be **paired** to securely communicate with each other.

4. Operates in Piconets:

One master device can connect with up to **7 active slave devices**.

5. Versions:

Different versions like **Bluetooth 4.0 (Low Energy)**, **5.0**, **5.2**, offering improved **speed, range, and efficiency**.

Advantages of Bluetooth:

Wireless Convenience: No need for cables to connect devices.
Low Cost and Easy Setup: Inexpensive and simple to configure.
Widely Supported: Built into most modern smartphones, laptops, and accessories.
Secure Connection: Offers encryption and frequency hopping to reduce interference.

Disadvantages of Bluetooth:

Limited Range: Effective only over short distances.
Lower Data Speed: Slower than Wi-Fi or other wireless technologies.
Interference: Can face interference from other wireless devices in the same frequency band.

In Summary:

Bluetooth is a **reliable and low-power** wireless technology used for **short-range communication** between devices, making it ideal for **hands-free calling, audio streaming, file sharing**, and **IoT applications**.

Unit III

3. Internet Basics and Networking Protocol

3.1 Internet- Architecture, Internet Service Providers (ISP), Internet Addressing System: IP Address, DNS, URL

3.2 Concept of Intranet & Extranet

3.3 Networking protocol: IP, TCP, FTP, HTTP, DHCP

3.1 Internet- Architecture, Internet Service Providers (ISP), Internet Addressing System: IP Address, DNS, URL:

1. Internet Architecture:

The Internet is a **global network of interconnected computer networks** based on the **TCP/IP protocol suite**.

It consists of:

Clients: Devices like computers, mobiles that request services.
Servers: Provide services like websites, email, files.
Routers & Switches: Direct data traffic between networks.
Backbone: High-speed data lines that connect major networks.
Protocols: Govern communication (e.g., HTTP, TCP/IP, FTP).

2. Internet Service Providers (ISPs):

ISPs are companies that **provide access to the Internet** to individuals and organizations.
Examples: **Jio, Airtel, BSNL, ACT, etc.**
They offer services like broadband, mobile data, domain registration, web hosting, and email.

3. Internet Addressing System:

a) IP Address (Internet Protocol Address):

A unique **numerical identifier** for each device on a network.
Two types:

IPv4 (e.g., 192.168.1.1) – 32-bit address.
IPv6 (e.g., 2001:0db8:85a3::8a2e:0370:7334) – 128-bit address.
Used for **identifying and locating devices**.

b) DNS (Domain Name System):

Converts **human-readable domain names** (like google.com) into IP addresses.
Acts like a **phonebook of the Internet**.
Example: `google.com` → `142.250.64.78`

c) URL (Uniform Resource Locator):

The **web address** used to access resources on the Internet.
Format: `https://www.example.com/page.html`

`https` = protocol
`www.example.com` = domain name
`/page.html` = path to resource

In Summary:

The Internet relies on a layered architecture, ISPs for access, and an addressing system using **IP, DNS, and**

URLs to connect users to services and resources across the world.

3.2 Concept of Intranet & Extranet:

1. Intranet:

An **Intranet** is a **private network** used **within an organization** to share information, resources, and services among employees.

Features of Intranet:

Accessible **only to authorized users** within the organization.
Uses **Internet technologies** (like web browsers, servers, emails).
Improves **internal communication** and **collaboration**.

Examples:

Company HR portals, internal employee dashboards, internal document sharing systems.

Advantages:

Enhances productivity and communication.
Secure internal information sharing.
Centralized access to organizational resources.

2. Extranet:

An **Extranet** is an extension of the intranet that provides **controlled access to external users** such as **partners, suppliers, or customers**.

Features of Extranet:

Combines **intranet and internet technologies**.
Access is granted through **secure login systems** (VPN, firewalls, etc.).
Enables **B2B (business-to-business)** communication.

Examples:

Vendor portals, partner collaboration platforms, customer order tracking systems.

Advantages:

Strengthens external partnerships.
Enhances supply chain management.
Secure sharing of specific information with outsiders.

In Summary:

Intranet is used for **internal communication** within an organization.
Extranet allows **secure external access** to selected parts of the internal network, improving collaboration with partners and clients.

3.3 Networking protocol: IP, TCP, FTP, HTTP, DHCP:

1. IP (Internet Protocol):

IP is responsible for **addressing and routing** data packets between devices on a network.
Every device is assigned a **unique IP address**.

It ensures data is sent to the correct destination.
Types: **IPv4** and **IPv6**.

2. TCP (Transmission Control Protocol):

TCP ensures **reliable and error-free communication** between devices.
It breaks data into packets, sends them, and ensures they are **delivered in order**.
Commonly used with IP (TCP/IP).

3. FTP (File Transfer Protocol):

FTP is used to **transfer files** between computers over a network.
Allows uploading and downloading files from servers.
Requires **username and password** for access.
Example: Uploading website files to a hosting server.

4. HTTP (HyperText Transfer Protocol):

HTTP is used for **transmitting web pages** on the Internet.
It enables **communication between web browsers and web servers**.
Example: Accessing a website like `http://www.example.com`.

5. DHCP (Dynamic Host Configuration Protocol):

DHCP **automatically assigns IP addresses** to devices on a network.
Eliminates the need for manual IP configuration.
Ensures that no two devices have the **same IP address** (avoids conflict).

In Summary:

These protocols handle key networking tasks such as **data transmission (TCP/IP)**, **file sharing (FTP)**, **web access (HTTP)**, and **automatic IP assignment (DHCP)**, forming the backbone of communication on the Internet and local networks.

Unit IV

4. Network Security

4.1 Network Security issues
4.2 Traditional Cryptography- substitution Ciphers, Transposition Ciphers
4.3 Two fundamental cryptographic principles
4.4 DES
4.5 Digital Signature
4.6 Firewalls

4.1 Network Security issues:

What is Network Security?

Network security refers to the **measures taken to protect data**, devices, and systems in a network from **unauthorized access, misuse, and attacks**.

Common Network Security Issues:

1. Unauthorized Access:

When an outsider gains access to a network or system without permission.
Can lead to **data theft or damage**.

2. Malware Attacks (Viruses, Worms, Trojans):

Malicious software that can **infect and damage systems**, steal data, or disrupt services.

3. Phishing and Social Engineering:

Tricks users into **revealing sensitive information** like passwords or financial details through fake emails or websites.

4. Denial of Service (DoS) Attacks:

Overloads a server or network with traffic, causing it to crash or become unavailable.

5. Data Interception (Sniffing/Eavesdropping):

Hackers can **intercept and read** unencrypted data being transmitted over a network.

6. Weak Passwords and Authentication:

Using **simple or reused passwords** makes it easy for attackers to guess and gain access.

7. Man-in-the-Middle (MITM) Attacks:

An attacker secretly intercepts and possibly alters communication between two parties.

In Summary:

Network security issues involve threats like **unauthorized access, malware, data theft, and service disruption**. To prevent these, networks must implement strong **encryption, firewalls, antivirus, access controls, and user awareness** programs.

4.2 Traditional Cryptography- substitution Ciphers, Transposition Ciphers:

What is Traditional Cryptography?

Traditional cryptography refers to **early methods of encrypting messages** to ensure **confidentiality**.
It involves converting plaintext into ciphertext using manual or simple algorithms, mainly categorized into:

1. Substitution Ciphers:

In substitution ciphers, **each letter or group of letters** in the plaintext is **replaced by another letter or symbol**.

Types:

Caesar Cipher:

Each letter is shifted by a fixed number of positions.
Example: Shift by 3 → A becomes D, B becomes E.

Monoalphabetic Cipher:

Each plaintext letter is mapped to a different ciphertext letter using a fixed key.

Advantages:

Simple to implement.
Easy to understand.

Disadvantages:

Vulnerable to **frequency analysis** attacks.

2. Transposition Ciphers:

In transposition ciphers, the **positions of the characters** in the plaintext are **rearranged** without changing the actual characters.

Example:

Plaintext: **HELLO**
Rearranged (e.g., reverse): **OLLEH**

Columnar Transposition:

Text is written in rows under a set of columns, and the columns are reordered based on a key.

Advantages:

Harder to break than simple substitution.
Preserves letter frequencies, which makes analysis harder.

Disadvantages:

Still vulnerable to pattern recognition.
Requires more complex keys for better security.

In Summary:

Traditional cryptography uses **substitution** (changing characters) and **transposition** (rearranging characters) techniques. Though basic, they laid the foundation for **modern encryption methods**.

4.3 Two fundamental cryptographic principles:

Cryptography is based on two fundamental principles that ensure **secure communication** over insecure channels:

1. Confusion:

Confusion means making the **relationship between the ciphertext and the encryption key as complex as possible**.

The idea is to prevent attackers from being able to **guess the key** even if they have some knowledge of the plaintext and ciphertext.

How it works:

Achieved using complex substitution techniques.

Each character in the plaintext is replaced unpredictably.

Example:

Substitution ciphers (e.g., Caesar Cipher) introduce confusion by replacing letters.

2. Diffusion:

Diffusion means **spreading the influence of a single plaintext character over many ciphertext characters**.

This helps hide **patterns** in the plaintext to make **statistical analysis more difficult**.

How it works:

Achieved using transposition techniques or mixing operations (in modern ciphers).

A small change in the plaintext should result in a major, unpredictable change in the ciphertext.

Example:

In block ciphers, changing one bit of the input should change many bits of the output.

In Summary:

Confusion hides the relationship between the key and ciphertext.

Diffusion hides patterns in the plaintext.

Together, these principles form the **foundation of secure encryption** systems.

4.4 DES:

What is DES?

DES (Data Encryption Standard) is a **symmetric key block cipher** developed by IBM and adopted by the U.S. government in 1977 for secure data encryption. It uses the **same key for encryption and decryption**.

Key Features of DES:

1. Block Size:

DES operates on **64-bit blocks** of plaintext and ciphertext.

2. Key Size:

Uses a **56-bit key** (plus 8 bits for parity = 64 bits total).

The 56-bit key was strong initially but is now **considered weak** due to brute-force vulnerabilities.

3. Number of Rounds:

DES performs **16 rounds** of complex operations including **substitution and permutation**.

4. Feistel Structure:

DES uses the **Feistel network**, where each round involves splitting the data into two halves and processing them.

5. Encryption & Decryption:

The same algorithm is used for both encryption and decryption, with the **key schedule reversed** for decryption.

Advantages:

Was widely used and trusted for decades.

Provided a standard method for encryption across platforms.

Disadvantages:

The **56-bit key is no longer secure** against modern brute-force attacks.

Replaced by **stronger algorithms** like **AES**.

In Summary:

DES was an early and influential **symmetric encryption algorithm** that played a major role in data security, but due to its **limited key size**, it has been **phased out in favor of more secure algorithms**.

4.5 Digital Signature:

What is a Digital Signature?

A **Digital Signature** is a **cryptographic technique** used to **verify the authenticity, integrity, and origin** of digital messages or documents. It acts like an **electronic fingerprint** of the sender.

Key Features of Digital Signatures:

1. Authentication:

Confirms that the message is sent by the **genuine sender**.

2. Integrity:

Ensures the message has **not been altered** during transmission.

3. Non-repudiation:

The sender **cannot deny** having sent the message, as the signature is unique and verifiable.

How It Works:

1. The sender uses their **private key** to generate a digital signature from the message using a **hash function**.
2. The signature is sent along with the original message.
3. The receiver uses the sender's **public key** to verify the signature and ensure that the message is authentic and unmodified.

Example Algorithms:

RSA, **DSA (Digital Signature Algorithm)**, and **ECDSA (Elliptic Curve DSA)**.

Advantages:

Provides **security** in digital communications.
Ensures **trust** in online transactions, emails, and legal documents.
Widely used in **e-Governance, banking**, and **e-commerce**.

In Summary:

A digital signature is a secure, electronic method of verifying the **origin, authenticity, and integrity** of digital data, playing a vital role in **secure communication and digital transactions**.

4.6 Firewalls:

What is a Firewall?

A **firewall** is a **network security system** (hardware or software) that **monitors and controls** incoming and outgoing network traffic based on **predefined security rules**. It acts as a **barrier between a trusted internal network and untrusted external networks**, such as the Internet.

Functions of a Firewall:

1. Packet Filtering:

Inspects data packets and allows or blocks them based on source/destination IP, port number, and protocol.

2. Access Control:

Restricts unauthorized users from accessing internal networks.

3. Monitoring and Logging:

Keeps records of traffic patterns and detects suspicious activity.

4. Protection Against Attacks:

Helps prevent **unauthorized access, malware, and hacking attempts**.

Types of Firewalls:

1. Hardware Firewall:

A physical device placed between the network and gateway.
Suitable for organizations.

2. Software Firewall:

Installed on individual computers or servers.
Ideal for personal or small office use.

3. Proxy Firewall:

Acts as an intermediary between users and the internet, providing additional filtering.

Advantages:

Enhances **network security**.

Controls access to **sensitive data**.

Helps in **preventing cyber attacks** like worms, viruses, and intrusions.

In Summary:

A firewall is a critical component of cybersecurity, ensuring that **only safe and authorized data** enters or leaves a network, thereby **protecting systems from external threats**.